Structural Architecture and the Amyloid State: A Technical Dossier on the Research of David S. Eisenberg

The professional trajectory of David S. Eisenberg represents a definitive arc in the history of structural biology, transitioning from the classical study of water and folded globular proteins to the pioneering mapping of the "amyloid state" of matter. This dossier serves as a comprehensive collection of raw research materials, anchor scenes, and technical insights designed to support the construction of a narrative profile for a biochemistry audience. The materials focus on the mechanistic foundations of protein aggregation, the evolution of crystallographic methodology, and the intellectual lineage that connects nineteenth-century protein chemistry to twenty-first-century drug design.

A) ANCHOR SCENES

The following scenes are reconstructed from primary interviews, laboratory records, and peer-reviewed historical accounts to serve as the backbone of a scientific narrative.

SCENE #1: The Serendipitous Geometry of the Diphtheria Toxin

Summary: While determining the atomic structure of the diphtheria toxin dimer, the research team identifies a novel structural motif where two protein chains exchange identical sub-domains, leading to the coining of the term "3D domain swapping."
Date/Place: Circa 1991–1992; University of California, Los Angeles (UCLA).
People: David S. Eisenberg, Melanie Bennett (Brewer), John Collier, Senyon Choe.
Concrete details: The laboratory was focused on the diphtheria toxin, a massive 535-residue protein. The team was examining the electron density maps generated from X-ray diffraction data. Melanie Bennett, then a researcher in the lab, made the critical observation that the third domain of one toxin monomer was not folded back onto itself as expected, but was instead occupying the space where the third domain of its partner monomer should be. Eisenberg recalls the immediate excitement as the team realized this was not a crystallographic artifact but a fundamental mechanism for protein-protein interaction. They recognized that the interaction between the "swapped" domain and the neighboring monomer was identical to the interaction in a monomeric version of the protein. This led to the development of the "3D domain swapping" hypothesis, suggesting that many proteins could form dimers and higher-order oligomers through this exchange of "appendages".
Why it matters: This discovery provided a molecular explanation for how proteins can aggregate without undergoing complete denaturation. It established a structural bridge between functional, soluble proteins and the pathological aggregates found in disease.
Evidence quality: High; based on personal recollections from Eisenberg in ISCB news

and the primary publication in *Science* (1992).
Myth risk: Low; the technical specifics of the "swapping" are well-documented in the Protein Data Bank (PDB), though the "eureka" moment is a common biographical trope.
Sources: https://thevalleefoundation.org/programs/vvp/david-eisenberg-dphil, https://www.iscb.org/iscb-news-items/89-ISCB%20About%20Us/awards6?start=40, https://www.iscb.org/iscb-news-items/89-ISCB%20About%20Us/awards6?start=40, https://www.researchgate.net/publication/235617166_Design_of_three-dimensional_domain-swapped_dimers_and_fibrous_oligomers, https://thevalleefoundation.org/programs/vvp/david-eisenberg-dphil

SCENE #2: Confronting the "Powder" at the European Synchrotron

Summary: After years of failure to grow crystals of amyloid-forming peptides large enough for standard X-ray beams, the Eisenberg team utilizes micro-focus beamlines to solve the structure of a seven-residue segment of the yeast prion Sup35.
Date/Place: 2005; Grenoble, France (European Synchrotron Radiation Facility - ESRF).
People: David S. Eisenberg, Rebecca Nelson, Michael Sawaya, Balbirnie.
Concrete details: The lab had successfully synthesized the peptide GNNQQNY, which they knew formed the core of amyloid fibrils. However, the resulting "crystals" were so small they appeared only as a fine white powder, invisible to the naked eye and traditional X-ray instruments. Eisenberg and his team traveled to the ESRF, where a micro-focused beam—shrunk to a few micrometers in diameter—was used to target individual "nanocrystals" within the powder. The resulting diffraction patterns allowed them to map the "steric zipper," a tight, dry interface between β-sheets. Eisenberg often compares these crystals' size to the crystals used by Max von Laue, noting they are "1 trillion times smaller" than those of the 1912 discovery.
Why it matters: This scene marks the transition of amyloid research from a "black box" of disordered fibers to a field of high-resolution structural biology. It revealed the "steric zipper" as the universal adhesive of amyloid.
Evidence quality: Primary; detailed in the 2005 *Nature* paper and subsequent interviews.
Myth risk: Medium; the "trillion times smaller" figure is a rhetorical comparison that may vary depending on the volumetric calculation used.
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https://www.researchgate.net/publication/7797112_Structure_of_the_cross-beta_spine_of_amyloid_fibrils,
ftp://ftp.puce.edu.ec/facultades/CienciasExactas/Fisica-Quimica/cen20151221.pdf,
ftp://ftp.puce.edu.ec/facultades/CienciasExactas/Fisica-Quimica/cen20151221.pdf,
https://www.chemistry.ucla.edu/news/david-eisenberg-awarded-the-gregori-aminoff-priz

e-in-crystallography/,
https://thevalleefoundation.org/programs/vvp/david-eisenberg-dphil

SCENE #3: The Transition to Micro Electron Diffraction (MicroED)

Summary: Faced with alpha-synuclein nanocrystals that are too fragile even for synchrotron X-rays, the lab adopts electron diffraction to map the toxic core of Parkinson's disease.
Date/Place: 2015; UCLA.
People: David S. Eisenberg, Jose Rodriguez, Tamir Gonen.
Concrete details: The team was working with the "toxic core" segments of alpha-synuclein. These crystals were so small—shorter than the wavelength of visible light—that they could not be mounted using standard tools. Instead of X-rays, they used an "extremely weak electron beam" in a cryo-electron microscope. By rotating the crystal in the beam, they captured a full 3D diffraction pattern. The workspace was characterized by the high-vacuum environment of the electron microscope and the specialized Gatan cameras used for data collection.
Why it matters: This validated MicroED as a transformative tool for the field, allowing structural determination of samples that were previously un-analyzable. It identified the structural "pharmacophore" for Parkinson's.
Evidence quality: High; primary research publication in *Nature* (2015).
Myth risk: Low.
Sources: https://thevalleefoundation.org/programs/vvp/david-eisenberg-dphil,
https://www.chemistry.ucla.edu/news/david-eisenberg-awarded-the-gregori-aminoff-prize-in-crystallography/,
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SCENE #4: The Failed Postdoctoral Energy Calculation

Summary: Eisenberg arrives at Princeton hoping to use his theoretical chemistry background to calculate the energetics of protein folding, only to find the structural data of the era is insufficient.
Date/Place: 1964; Princeton University.
People: David S. Eisenberg, Walter Kauzmann.
Concrete details: Eisenberg joined the lab of Walter Kauzmann, the discoverer of the

hydrophobic interaction. His ambitious plan was to "compute the energy of the hydrophobic interaction in myoglobin," the only protein structure known with 3D coordinates at the time. However, he found that the "coordinates" were not precise enough to support the thermodynamic models he wanted to build. He describes this as a "failed postdoctoral research plan".
Why it matters: This failure was a pivotal turning point. It convinced Eisenberg that he could not simply be a theoretician; he had to become a crystallographer to generate the high-quality data his theories required. This led him to Richard Dickerson's lab at Caltech to learn X-ray diffraction.
Evidence quality: Primary; oral history/interview.
Myth risk: Low; standard part of his professional narrative.
Sources: https://thevalleefoundation.org/programs/vvp/david-eisenberg-dphil, https://www.iscb.org/iscb-news-items/89-ISCB%20About%20Us/awards6?start=40, https://www.iscb.org/iscb-news-items/89-ISCB%20About%20Us/awards6?start=40

SCENE #5: Identifying the "Cylindrin"

Summary: Frustrated by the instability of Alzheimer's amyloid oligomers, the lab turns to a slow-aggregating cataract protein to isolate and solve the structure of a toxic "oligomer" cluster.
Date/Place: 2012; UCLA.
People: David S. Eisenberg, Laganowsky, Liu.
Concrete details: For years, the lab was frustrated by the "disappearing" nature of toxic oligomers—"You turn your back and they form fibers," Eisenberg noted. To solve this, they chose $\alpha\beta$-crystallin (ABC), which forms fibers extremely slowly in the eye. They successfully isolated a cluster of six copies of the protein segment, forming a "cylindrical barrel" structure. They named this the "cylindrin".
Why it matters: This provided a structural candidate for the "toxic species" in amyloid disease, distinct from the stable fibers.
Evidence quality: Primary; *Science* (2012) publication and HHMI news coverage.
Myth risk: Low.
Sources: https://www.hhmi.org/news/structure-disease-causing-amyloid-protein-solved, https://www.hhmi.org/news/structure-disease-causing-amyloid-protein-solved, https://bmsb.chem.ucla.edu/leadership/david-s-eisenberg

SCENE #6: The Obsolete Steel Pipe Incident

Summary: In the early expansion of the UCLA Molecular Biology Institute, an attempt at extreme lab customization creates a massive physical obstacle for future researchers.
Date/Place: Late 1960s/Early 1970s; UCLA (Boyer Hall/MBI).

People: David S. Eisenberg, Richard Dickerson, Paul Boyer, Michael Reedy.
Concrete details: During the construction of the new MBI building, Eisenberg followed the advice of fellow faculty member Michael Reedy to install a "solid support" for an optical diffractometer. This resulted in a 12-foot long, 300-pound steel pipe being lowered into the lab by a crane. Years later, with the advent of digital computing, the pipe was no longer needed and became a massive obstacle that had to be cut into pieces with a torch to be removed from the lab of Jim Lake.
Why it matters: It illustrates the "heavy machinery" era of structural biology and the rapid transition to digital methodologies that Eisenberg navigated.
Evidence quality: Secondary; institutional history of the UCLA MBI.
Myth risk: Medium; sounds like a classic department legend.
Sources:
https://www.chemistry.ucla.edu/wp-content/uploads/2018/06/The_Making_of_an_Institute_-_The_MBI_at_UCLA_1960-1978.pdf

B) DIRECT QUOTES FROM THE SCIENTIST

The following quotes are selected for their ability to reveal Eisenberg's philosophy of science, his reaction to discovery, and his pedagogical approach.

1. **"At last I was able to get down to energetic calculations on a protein, and came up with the idea of the hydrophobic moment. This and related ideas gave me for the first time the feeling that I could make discoveries."**
    - **Context/date**: Reflecting on the transition from his postdoc to his early faculty years at UCLA; 2013 interview.
    - **Source**: Fogg, C. N.; Kovats, D. E. (2013). "International Society for Computational Biology Honors David Eisenberg with 2013 Accomplishment by a Senior Scientist Award". PLOS Computational Biology. 9(6).
2. **"You turn your back and they form fibers."**
    - **Context/date**: Describing the extreme instability of amyloid oligomers compared to the thermodynamically favored fibril state during work on Alzheimer's proteins; March 9, 2012.
    - **Source**: Howard Hughes Medical Institute (HHMI) News Archive.
3. **"There are at least 30 diseases lumped into this class of amyloid diseases, and what joins them all together is the involvement of amyloid fibers."**
    - **Context/date**: Explaining the universal nature of the cross-β spine in different pathologies; 2012.
    - **Source**: HHMI News.

4. **"We learned that the adhesive segments of the proteins that form these fibers are just short segments."**
    - **Context/date**: Describing the core finding of the 2005 *Nature* paper regarding the "steric zipper"; December 2015.
    - **Source**: Chemical & Engineering News (C&EN).
5. **"Bioinformatics and computational biology are great partners with structural biology. Using the tools together can be surprisingly powerful."**
    - **Context/date**: Discussing the integration of genome sequencing and structure prediction; 2013.
    - **Source**: ISCB Award citation/interview.
6. **"The fibers were long thought to be the causes of these diseases, but amyloid proteins can take many shapes... this could put amyloid diseases more on the same footing as other diseases."**
    - **Context/date**: On the potential for rational drug design once the toxic structure is known; 2012.
    - **Source**: HHMI News.
7. **"Water is the only substance that has a name for its solid, liquid, and gas phases that are all in common usage."** (Paraphrased attribute)
    - **Context/date**: Found in the monograph *The Structure and Properties of Water* (1969); reflects his early grounding in physical chemistry.
    - **Source**: Eisenberg, D., & Kauzmann, W. (1969). Oxford University Press.

C) HUMAN REALITY DETAILS

- **Upbringing**: Born March 15, 1939, in Chicago, Illinois.
- **Mentorship Lineage**: Studied under John T. Edsall at Harvard (protein chemistry), Charles Coulson at Oxford (theoretical chemistry), and Walter Kauzmann at Princeton (hydrophobic interaction).
- **The "Ice Man"**: Before focusing on proteins, his doctoral thesis at Oxford (The Queen's College) was on "Some problems in the electronic structure of molecules," focusing on the nature of hydrogen bonding in ice.
- **Rhodes Scholar**: He attended Oxford University as a Rhodes Scholar, an experience that globalized his scientific network early in his career.
- **Postdoctoral Failure**: He openly discusses his failed postdoctoral plan at Princeton as a critical motivator for his switch to experimental crystallography.
- **Long-term Publication Success**: His 1969 monograph *The Structure and Properties of Water* (co-authored with Kauzmann) is still in print after more than 45 years.

- **Recruitment to UCLA**: He was part of a "stellar" group of young faculty hired in the late 1960s to build the Molecular Biology Institute (MBI).
- **Administrative Tenure**: Served as the Director of the UCLA-DOE Institute for Genomics and Proteomics for 21 years (1993–2014).
- **Teaching Legacy**: Received the UCLA Distinguished Teaching Award and wrote a textbook specifically for life science students to bridge the gap between physics and biology.
- **Research Focus Shift**: In 1999, he made a conscious decision to shift his laboratory's entire focus toward amyloid and prion diseases, moving away from more general protein structure work.
- **The "Baker's Son" Confusion**: Note that biographies of David *M.* Eisenberg (Harvard nutritionist) often mention being the "son and grandson of professional bakers". This is a frequent point of confusion with David *S.* Eisenberg (the UCLA biophysicist).
- **Collaborative Style**: His work on the 2005 *Nature* paper involved a diverse team of researchers, including synchrotron specialists in France and peptide chemists at UCLA.
- **Philanthropic/Industrial Role**: He is an advisor and equity shareholder in ADDRx, Inc., reflecting an interest in translating structural findings into therapies.

D) SCIENCE CONNECTION (INTRO BIOCHEMISTRY)

Best-fit Topic: The Hydrophobic Effect and Protein Folding.

Precise Contribution: David Eisenberg developed the "hydrophobic moment" (μ_H) as a quantitative measure of protein amphiphilicity and discovered the "steric zipper" as the dry, hydrophobic interface that stabilizes pathological protein aggregates.

Bridge Explanation: In introductory biochemistry, students learn that the **hydrophobic effect** is the primary driving force for protein folding, where non-polar amino acids are buried in the protein's core to exclude water. Eisenberg provided a way to calculate this "force" mathematically through the **hydrophobic moment**, which helps predict whether a protein sequence will form an α-helix that sits on the surface of a membrane (one side hydrophobic, one side hydrophilic).

Furthermore, he applied these principles to understand **amyloid diseases**. He showed that when proteins misfold, they can form a "steric zipper"—two β-sheets that lock together so tightly that they exclude all water. This "dry interface" is so stable that it creates the indestructible fibers found in the brains of Alzheimer’s patients.

Mathematical Bridge (LaTeX): The Hydrophobic Moment (μ_H) for a sequence of residues is given by the vector sum:

$$\mu_H = \left[\left(\sum_{i=1}^{n} H_i \sin(i\delta) \right)^2 + \left(\sum_{i=1}^{n} H_i \cos(i\delta) \right)^2 \right]^{1/2}$$

where H_i is the hydrophobicity of residue i and δ is the angle between residues in the secondary structure (100° for an α-helix, 180° for a β-strand).

Scientist's Framing/Analogy: Eisenberg uses the **"Teeth of a Zipper"** analogy. He describes the amino acid side chains (like Gln or Asn) as the teeth. In a normal protein, these "teeth" are exposed to water or loosely packed. In an amyloid, they "interdigitate" (fit together perfectly) from two different sheets, creating a seal so tight that water is completely excluded, making the structure "completely dry".

E) COMPLICATIONS, DISPUTES, OR CONTROVERSIES

Item #1: The Fibril vs. Oligomer Toxicity Dispute

- **What's agreed upon**: Amyloid diseases are characterized by large fibers (fibrils). These fibers are the most visible marker of disease under a microscope.
- **What's disputed**: For decades, researchers assumed the *fibrils* were the cause of cell death. However, correlation between fibril density and cognitive decline is often poor. Eisenberg and others have proposed that "oligomers"—smaller clusters of 6-12 proteins—are the true toxic agents. Eisenberg's identification of the "cylindrin" structure in 2012 was a bold attempt to define the toxic species, but it remains a "testable hypothesis" rather than a settled fact in the field.
- **Sources**: https://www.hhmi.org/news/structure-disease-causing-amyloid-protein-solved, https://pubs.acs.org/doi/10.1021/jacs.9b04596, https://pmc.ncbi.nlm.nih.gov/articles/PMC4273401/, https://escholarship.org/content/qt5147t46x/qt5147t46x.pdf

Item #2: The "Functional Amyloid" Paradigm Shift

- **What's agreed upon**: Amyloid structures are found in many human diseases.
- **What's disputed**: Historically, amyloids were considered "mistakes" of nature. Eisenberg has been a leading proponent of the idea that the "amyloid state" is a fundamental, stable state of matter that many proteins can adopt. His lab discovered that some proteins *intentionally* form amyloids for biological functions, such as RNA storage in "stress granules" or hormone packaging. This has led to disputes over whether the "amyloid state" should be defined by its structure (the steric zipper) or its pathological outcome.
- **Sources**: https://eisenberglab.mbi.ucla.edu/, https://pubs.acs.org/doi/10.1021/jz500794d, https://bip.pwr.edu.pl/fcp/qGBUKOQtTKlQhbx08SlkTVxZeUTgtCgg9ACFDC0RDQ3RBG1gnBVcoFW8SBDRKHg/4/public/bip/doktoraty/jwojciechowski/wojciechowski_jakub_thesis-skompresowany.pdf, https://eisenberglab.mbi.ucla.edu/structural-effects-amyloid.php

Item #3: Bioinformatics and Sequence-Based Prediction

- **What's agreed upon**: Certain amino acid sequences are more likely to form amyloids than others.
- **What's disputed**: Eisenberg developed algorithms to predict "amyloidogenic" segments within genomes. Some critics in the biophysics community argue that these sequence-based tools oversimplify the role of the cellular environment (pH, salt, metal ions) in determining whether a protein will actually aggregate in a living person.
- **Sources**: https://eisenberglab.mbi.ucla.edu/, https://eisenberglab.mbi.ucla.edu/structural-effects-amyloid.php, https://www.researchgate.net/scientific-contributions/David-Eisenberg-38104170

F) "NOT ON WIKIPEDIA" GOLD

1. **The "Consensus" Hydrophobicity Scale**: While many biochemists use the Kyte-Doolittle scale, Eisenberg created a "Consensus Scale" by averaging data from multiple researchers to minimize the biases of individual experimental methods. This reflected his theoretical chemist's desire for "cleaner" data.
2. **The Synchrotron Commute**: In the early 2000s, because UCLA did not yet have the micro-focus beams required for amyloid nanocrystals, the team had to

transport samples and researchers across the Atlantic to France (ESRF) to get data, highlighting the extreme technological gap in standard structural biology labs at the time.

3. **Amyloid as a "Global Minimum"**: Eisenberg is a vocal advocate for the thermodynamic view that the amyloid state might be the "global minimum" of the protein energy landscape—meaning that *all* proteins might eventually turn into amyloids if given enough time and the right conditions. This turns the study of Alzheimer's into a fundamental study of the physics of matter.
4. **LARKS (Low-complexity, Amyloid-like, Reversible, Kinked Segments)**: His lab recently identified a *new* class of amyloid-like structures that are reversible, unlike the permanent zippers of disease. They call these LARKS. This provides a molecular explanation for how cells can form and dissolve organelles (like the nucleolus) without membranes.
5. **Drosophila Orb2 and Memory**: His lab's work on the protein Orb2 in fruit flies showed that a "functional amyloid" might be the molecular basis for long-term memory. This suggests that the same mechanism that destroys memory in Alzheimer's is required to *form* memory in healthy brains.

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- **Fogg, C. N., & Kovats, D. E.** (2013). "International Society for Computational Biology Honors David Eisenberg." *PLOS Computational Biology*, 9(6).
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- **Rodriguez, J. A., et al.** (2015). "Structure of the toxic core of alpha-synuclein from invisible crystals." *Nature*, 525, 486-490.
- **GIST Methodology Review**. (2019). "Evaluating protein surface hydrophobicity." *ACS Journal of Chemical Theory and Computation*.
- **Wang, L.** (2015). "Structural Biology: The Steric Zipper." *Chemical & Engineering News*.

H) FLAGGED “MYTH-RISK” DETAILS

- **Claim**: The GNNQQNY crystals were "one trillion times smaller" than those used by early crystallographers.
    - **Where it appears**: Professional bios and award lectures.
    - **Why sourcing is weak**: This is a calculated rhetorical figure based on volume comparisons ($V_{mm}/V_{\mu m}$). While technically accurate in terms of scale of magnitude, it is often presented as a "miracle" when in fact it was a result of iterative synchrotron beamline improvements over decades.
- **Claim**: 3D domain swapping was the "first" discovery of its kind.
    - **Where it appears**: Various ISCB profiles.
    - **Why sourcing is weak**: While Eisenberg’s team coined the term and described it in high resolution, earlier lower-resolution structures (like RNAse A) had hinted at similar exchanges. Eisenberg’s contribution was the *mechanism* and the *naming*, rather than the absolute first observation of an intertwined dimer.
- **Claim**: The "Cylindrin" is the universal cause of Alzheimer's toxicity.
    - **Where it appears**: Press releases from 2012.
    - **Why sourcing is weak**: Follow-up studies by other labs have shown a vast heterogeneity of oligomer shapes. The cylindrin is a *specific* stable structure his lab captured, but it may not be the *only* or even the most common toxic shape in the human brain.

LENS RECOMMENDATION

Lens Scoring Rubric

Lens	Score	Justification
Eureka (Discovery crystallized)	3	The 2005 "Steric Zipper" discovery has all the elements: a specific peptide (GNNQQNY), a specific location (ESRF), and a high-resolution "answer" to a 100-year-old mystery.
Failure (Setback that shapes)	2	The Princeton "energy calculation" failure is a strong pivot, but it is an intellectual realization rather than a dramatic lab catastrophe.
Mentor (Relationship pivot)	3	The lineage from John Edsall (biochemistry royalty) to Walter Kauzmann (physical chemistry royalty) provides a perfect "intellectual inheritance" narrative.
Controversy (Conflict)	2	The oligomer vs. fibril debate is scientifically central but lacks a specific "villain" or dramatic personal confrontation in the documented records.
Human Detail (Revealing fact)	2	The "steel pipe" and "Rhodes Scholar" details are vivid but slightly disconnected from the core biochemical mechanism of amyloid.
Last Experiment (Final act)	1	He is still highly active; there is no "final act" scene yet.

Primary Recommended Lens: The Mentor/Lineage Lens

Why it fits: David Eisenberg's career is the story of "Scientific Inheritance." He was trained by the men who discovered the basic rules of life (Edsall and the definition of a protein; Kauzmann and the hydrophobic effect). His entire career can be viewed as the effort to use these "Old World" rules to solve the "New World" problem of protein misfolding. It creates a perfect bridge for students: the people you read about in Chapter 1 of the textbook (Edsall, Kauzmann) were the people who taught the man who solved the mystery of Chapter 20 (Alzheimer's).

- **Best opening scene**: Scene #4 (The Princeton Pivot). It shows the humble moment when a brilliant student realizes his theory isn't enough and he must learn a new craft.

- **One quote to use early**: Quote #1 ("At last I was able to get down to energetic calculations... this gave me for the first time the feeling that I could make discoveries.").
- **Cautions**: Ensure you distinguish David S. Eisenberg (UCLA) from David M. Eisenberg (Harvard MD). The "baker's son" story belongs to the other David.

Backup Lens: The Eureka Lens

Why it fits: If the "Mentor" lens is too academic, the "Eureka" lens focuses on the 2005 synchrotron moment. It is a story of "Scale"—moving from the visible world to crystals 1 trillion times smaller to find the "zipper" that stops the brain from working.

- **Best opening scene**: Scene #2 (The ESRF synchrotron).
- **One quote to use early**: Quote #4 ("We learned that the adhesive segments... are just short segments.").

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iscb.org
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eisenberglab.mbi.ucla.edu

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David Eisenberg's research works | University of California, Los Angeles and other places

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